

SURA, Russian Federation

WMO: 226760

Lat: 63.583N

Lon: 45.633E

Elev: 62

StdP: 100.58

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.9	(c) -32.9	(d) -39.3	(e) 0.1	(f) -36.8	(g) -35.3	(h) 0.1	(i) -32.8	(j) 8.5	(k) -6.9	(l) 7.8	(m) -6.7	(n) 0.7	(o) 160	(p) 0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	27.9	18.5	25.8	17.7	23.8	16.6	20.0	25.4	18.9	23.9	17.7	22.1	3.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	22.0	17.0	12.2	20.9	15.8	11.3	20.1	57.4	25.5	53.9	24.0	50.3	22.2	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.0	7.1	6.3	-38.6	30.3	4.1	1.9	-41.6	31.7	-44.0	32.8	-46.3	33.9	-49.3	35.3
			-38.7	21.4	4.1	1.6	-41.7	22.6	-44.1	23.5	-46.4	24.4	-49.3	25.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.9	-14.2	-13.4	-7.0	0.5	7.4	12.8	16.2	13.5	8.2	1.7	-5.6	-10.7		
	DBStd	12.18	9.47	9.28	6.30	5.10	5.29	4.67	4.15	4.03	3.68	4.02	6.89	9.01		
	HDD10.0	3816	750	656	527	286	117	22	3	11	77	258	467	641		
	HDD18.3	6419	1008	889	785	534	340	176	91	159	304	516	717	899		
	CDD10.0	483	0	0	0	2	36	107	197	119	23	1	0	0		
	CDD18.3	45	0	0	0	0	1	9	26	8	0	0	0	0		
	CDH23.3	519	0	0	0	1	24	131	277	85	1	0	0	0		
	CDH26.7	111	0	0	0	0	4	30	59	18	0	0	0	0		
(14) Wind	WSAvg	2.9	2.7	2.9	3.1	3.1	3.3	3.0	2.7	2.7	2.8	3.2	2.9	2.9		
(15) Precipitation	PrecAvg	557	37	27	28	31	47	61	65	69	53	56	46	39		
	PrecMax	694	79	52	45	62	80	157	151	125	100	104	75	59		
	PrecMin	459	17	4	6	7	14	20	17	34	23	15	16	24		
	PrecStd	60	12	11	11	14	18	30	31	23	20	23	16	10		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	2.4	7.3	18.7	26.8	29.9	30.9	29.2	21.3	13.0	6.5	3.2		
		MCWB	0.6	1.6	2.5	10.0	15.5	18.7	20.2	20.4	15.2	9.9	5.9	2.4		
	2%	DB	0.3	1.0	4.0	13.0	22.6	26.9	28.5	25.6	18.4	10.2	4.6	1.5		
		MCWB	-0.1	0.5	1.4	6.8	13.2	17.2	19.1	18.4	13.8	8.4	3.9	0.7		
	5%	DB	-1.0	-0.1	2.5	9.6	20.0	24.3	26.5	23.0	16.0	8.7	3.0	0.8		
		MCWB	-1.4	-0.7	0.7	4.7	12.4	15.8	18.3	17.1	12.7	7.5	2.6	0.3		
	10%	DB	-2.4	-1.6	1.3	7.5	17.3	21.8	24.7	20.6	14.0	7.2	1.6	0.0		
		MCWB	-2.9	-2.1	0.0	3.9	10.7	14.8	17.5	16.1	11.7	6.3	1.1	-0.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.9	1.7	3.3	10.9	16.9	20.2	21.9	21.2	16.1	10.8	6.0	2.3		
		MCDB	1.0	2.2	5.3	18.2	22.3	27.2	28.2	26.9	19.7	12.7	6.5	3.1		
	2%	WB	0.0	0.5	2.0	7.5	14.7	18.6	20.4	19.3	14.5	8.8	4.0	0.9		
		MCDB	0.3	0.9	3.6	11.6	21.0	24.8	25.9	24.1	17.3	9.8	4.5	1.2		
	5%	WB	-1.3	-0.6	1.0	5.6	12.9	17.2	19.4	17.8	13.1	7.6	2.6	0.4		
		MCDB	-1.0	-0.2	2.1	8.8	19.0	22.3	24.4	21.4	15.4	8.6	3.0	0.8		
	10%	WB	-2.9	-2.1	0.2	4.1	11.2	15.6	18.4	16.6	11.9	6.3	1.2	-0.5		
		MCDB	-2.5	-1.6	1.3	6.9	16.3	20.4	23.1	19.9	13.9	7.2	1.5	0.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	8.2	9.7	9.2	10.2	10.3	10.5	8.9	7.3	4.2	4.5	5.8		
		MCDBR	5.0	4.4	8.6	13.0	15.3	14.5	14.0	13.3	11.5	6.4	3.7	3.7		
	5% WB	MCWBR	5.1	4.2	6.4	7.8	7.6	6.4	5.8	6.4	7.0	5.1	3.8	3.6		
		MCWBR	5.2	4.8	6.1	11.3	13.3	12.4	11.9	9.9	6.2	3.8	3.8	3.5		
(40) Clear-Sky Solar Irradiance	taub		0.231	0.285	0.314	0.351	0.350	0.351	0.392	0.365	0.333	0.307	0.237	0.217		
		taud	1.955	2.060	2.103	2.146	2.433	2.485	2.374	2.428	2.506	2.365	2.034	1.940		
	Ebn at Noon		343	614	760	817	864	870	815	807	766	616	336	152		
		Edh at Noon	31	68	98	118	98	95	104	90	68	51	28	14		
(44) All-Sky Solar Radiation	RadAvg		0.18	0.79	2.13	3.84	4.79	5.21	5.12	3.54	1.89	0.69	0.19	0.07		
	RadStd		0.02	0.17	0.32	0.44	0.34	0.52	0.43	0.31	0.22	0.10	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page